👩🏾‍🏫 Staff Guide · Small Engines Overhaul · Unit 4 Compression (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow the class teacher's lesson and slides with Level 3 readings, guided notes and a quick check.

1. How the pieces line up

Gen ed lesson	Level 3 piece
His readings and worksheets	Level 3 reading (5 short paragraphs) + check questions
Class notes	Level 3 guided notes + key
Unit quiz / test	Level 3 quick check + key
Review (Kahoot / Jeopardy)	1 Level 3 game on the class site: 10 questions + a matching round

2. Supports built in

- 🔊 Read aloud on the reading and notes pages (tap the button)
- Key terms listed at the top of the notes
- Short-answer lines sized for 1-2 sentences

3. What to grade

- Notes and reading checks: same points as his worksheets for that lesson
- Quick check: score, then scale to his test points
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Compression Article**: The article is a scanned image (source unknown); our reading covers the same ideas.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

💨 Compression: Guided Notes

Small Engines Overhaul · Unit 4 · Level 3 (closer to the class work) · goes with the Compression Handout and article

Name: _____

Use the class slides and your reading. Key words:

compression | *combustion chamber* | *piston rings* | *valve seat* | *head gasket* | *compression gauge* | *PSI* | *compression release*

A. Sealing

B. Diagnosis

2. A gauge reads low. After adding a spoon of oil, it rises a lot. What is worn?

_____

_____

3. If the reading stays low after adding oil, suspect the _____ or head gasket.

C. Test steps

4. Write the compression test steps in order.

_____

_____

_____

💨 Compression: Reading

Small Engines Overhaul · Unit 4 · Level 3 (closer to the class work) · goes with the Compression Handout and article

Name: _____

Compression: The Squeeze That Makes Power

Compression is the squeezing of the air-fuel mixture during the second stroke. When the mixture is squeezed into a small space and then ignited, it burns faster and builds more pressure, pushing the piston down with more force. Good compression is needed for easy starting and full power.

Several parts seal the combustion chamber. Piston rings seal the piston to the cylinder wall. The intake and exhaust valves must close tightly against their seats. The head gasket seals the cylinder head to the block, and the spark plug seals the plug hole.

When any seal fails, pressure leaks out. Common causes are worn or stuck rings, a scored cylinder, burned or carbon-covered valves, incorrect valve clearance and a blown head gasket.

A compression test checks the seal. With the spark plug removed, a compression gauge is screwed into the plug hole and the engine is turned over several times. The reading in PSI is compared to the manufacturer's specification. Many small engines have a compression release that lowers the reading, so always use the manual's spec.

Check your reading

1. Why does compression increase power?

2. Name three parts that seal the combustion chamber.

3. How is a compression test done?

📟 Compression Test: Lab Sheet

Small Engines Overhaul · Unit 4 · Level 3 · goes with Compression Handout

Name: _____

Question: Does our practice engine have good compression, and if not, where is the leak?

Materials

compression gauge, spark plug socket, oil can, engine, spec sheet

⚠️ Safety

- Safety glasses
- Plug wire disconnected
- Face away from the plug hole
- Tools with adult supervision

Prediction

Predict: will the dry reading be in spec?

_____

_____

Procedure

1. Remove plug; install gauge.
2. Dry test: 5 pulls, record.
3. Add 1 tsp oil.
4. Wet test: 5 pulls, record.
5. Compare to spec; diagnose.

Data

Test	PSI	Spec	In spec?
Dry			
Wet			

Analysis (Claim, Evidence, Reasoning)

1. Claim: where is the leak? Evidence: dry vs wet readings. Reasoning: what the oil does.

💨 Compression: Quick Check

Small Engines Overhaul · Unit 4 · Level 3 (closer to the class work) · goes with the Compression Handout and article

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Which part seals the piston to the cylinder wall?

A. Head gasket | B. Valve seat | C. Piston rings

2. Compression is measured in…

A. PSI | B. RPM | C. Volts

3. A compression release…

A. Adds fuel | B. Cools the engine | C. Makes the engine easier to pull-start

4. Low compression usually causes…

A. Hard starting and low power | B. Higher top speed | C. Louder spark

5. Name two causes of low compression.

6. Why compare a reading to the manual's spec?

💨 Compression: Guided Notes · ANSWER KEY

Small Engines Overhaul · Unit 4 · Level 3 (closer to the class work) · goes with the Compression Handout and article

Name: _____

Use the class slides and your reading. Key words:

compression | *combustion chamber* | *piston rings* | *valve seat* | *head gasket* | *compression gauge* | *PSI* | *compression release*

A. Sealing

B. Diagnosis

2. A gauge reads low. After adding a spoon of oil, it rises a lot. What is worn?
The piston rings (oil temporarily seals them).

3. If the reading stays low after adding oil, suspect the **valves** or head gasket.

C. Test steps

4. Write the compression test steps in order.
Remove plug, install gauge, crank 5 times, record PSI, compare to spec.

🔑 *Key written by our team from the class teacher's worksheets and slides, please check.*

📟 Compression Test: Lab Sheet · ANSWER KEY

Small Engines Overhaul · Unit 4 · Level 3 · goes with Compression Handout

Name: _____

Question: Does our practice engine have good compression, and if not, where is the leak?

Materials

compression gauge, spark plug socket, oil can, engine, spec sheet

⚠️ Safety

- Safety glasses
- Plug wire disconnected
- Face away from the plug hole
- Tools with adult supervision

Prediction

Predict: will the dry reading be in spec?

Answers vary.

Procedure

1. Remove plug; install gauge.
2. Dry test: 5 pulls, record.
3. Add 1 tsp oil.
4. Wet test: 5 pulls, record.
5. Compare to spec; diagnose.

Data

Test	PSI	Spec	In spec?
Dry	**45**	**60+**	**no**
Wet	**70**	**60+**	**yes**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: where is the leak? Evidence: dry vs wet readings. Reasoning: what the oil does.
Example: wet rose 25 PSI, so the rings are worn; oil sealed them temporarily.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

💨 Compression: Quick Check · ANSWER KEY

Small Engines Overhaul · Unit 4 · Level 3 (closer to the class work) · goes with the Compression Handout and article

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Which part seals the piston to the cylinder wall?

A. Head gasket | B. Valve seat | **C. Piston rings**

2. Compression is measured in…

A. PSI | B. RPM | C. Volts

3. A compression release…

A. Adds fuel | B. Cools the engine | **C. Makes the engine easier to pull-start**

4. Low compression usually causes…

A. Hard starting and low power | B. Higher top speed | C. Louder spark

5. Name two causes of low compression.

Worn rings, burned valves, blown head gasket, scored cylinder, wrong valve clearance.

6. Why compare a reading to the manual's spec?

Specs differ by engine and compression release; the spec tells you if it is normal.

🔑 *Written by our team from his unit calendar, worksheets and review games, please check.*